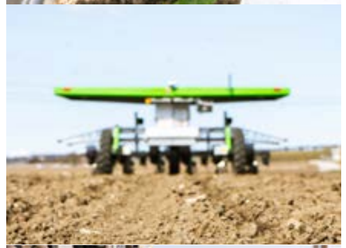
FARMDROID

AUTOMATIC SEEDING, WEEDING AND MICROSPRAYING ROBOT











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Dear Fellow Farmer,

We know firsthand the hurdles you face daily. The increasing difficulty in securing manual labor for organic farming, and the diminishing options for plant protection for conventional farms, are challenges we are intimately familiar with.

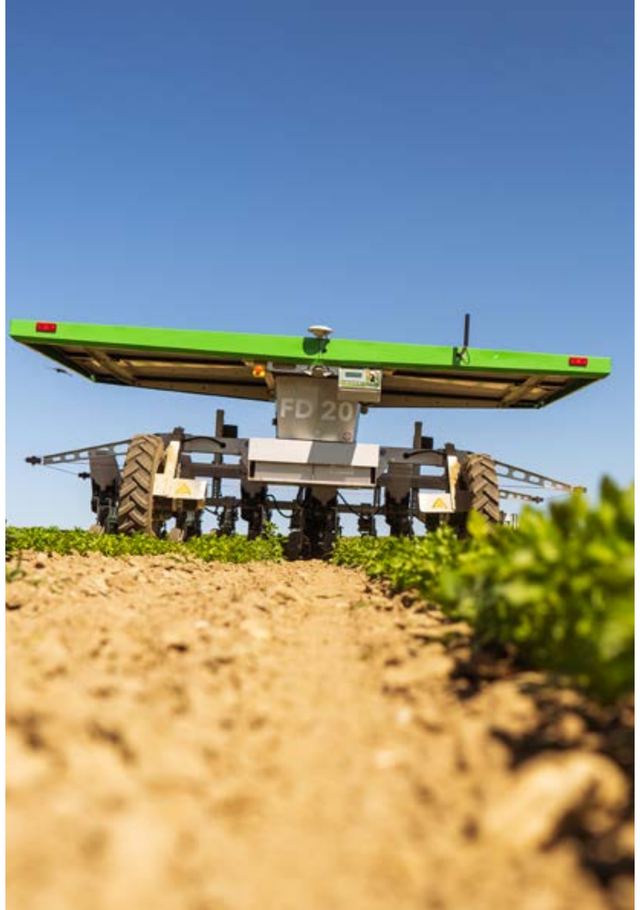
That is precisely why the FarmDroid FD20 was born. Designed by farmers, for farmers, we crafted this machine rooted in our shared experiences, the very challenges that we, as a farming community, face every day.

Our unwavering promise to you is this: in everything we do, our focus remains on making your life as a farmer easier, reducing your costs, and maximizing your yield.

Stepping into the realm of agricultural robotics might feel like venturing into the future. Yet, with us, that future has already dawned. With our network of trusted distributors spanning the globe, you are not just investing in a machine; you are joining a community. From the initial setup of the FD20 in your fields to unwavering support throughout each season, you will always have a trusted partner by your side.

At our core, we have the DNA of a farmer. We understand, we innovate, and most importantly, we are here for you. The culmination of our work is where cutting-edge technology meets dedication, blending robotic precision with a genuine human touch.

Warm regards, the FarmDroid Team

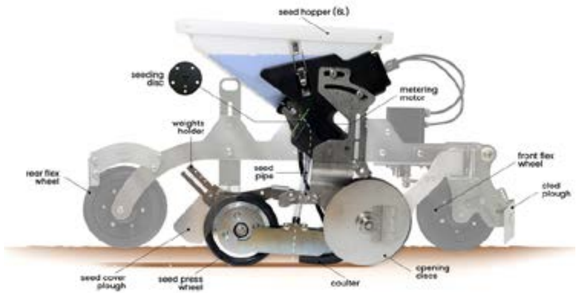


EXCEPTIONAL 8MM PRECISION SEEDING FOR OPTIMAL GERMINATION

Seed

At the core of FarmDroid FD20 is its advanced seeding system. Once set up, the robot plans the entire field layout, placing each seed with 8mm accuracy. The result? A uniform pattern across the field, ensuring every plant has the optimal space to flourish.

What differentiates the FarmDroid FD20 is its exceptional 8mm GPS precision. When the field is set up, the robot calculates where each seed will be placed, enabling it to commence weed control even before crops germinate — a unique capability that remains unmatched in the industry! This approach keeps fields weed free right from the start, promoting optimal germination and higher yields.





A Robot You Can Trust

Using FarmDroid FD20 is worry-free. The robot detects each seed it is sowing, and if a seed gets stuck or the robot runs out of seeds, it stops and sends a notification to your smartphone right away. This way, you are informed of the robot's process in real time.

Versatility is another hallmark of the FD20 seeding system. Suitable for over 50 different crops, and with adjustable row spacings from 22.5 cm to 90 cm, it's designed to cater to diverse seeding requirements.

Compatible Crops

As of today, the robot has already with great success taken care of many different crops, such as:

Parsley



Onion



Beetroot



Sugar beet



Coriander



Turnip



Rapeseed



Chicory



Flower seeds



- Arugula
- Cabbage
- Cauliflower
- Dill
- Hemp
- Kale
- Lettuce
- Radish
- Spinach

Seed Testing Lab

Our Seed Test Lab closely collaborates with farmers to experiment with new crops' compatibility with our robot's sowing system. Currently our robot can work with more than 50 different crops.





If We Can Seed It, We Can Weed It!

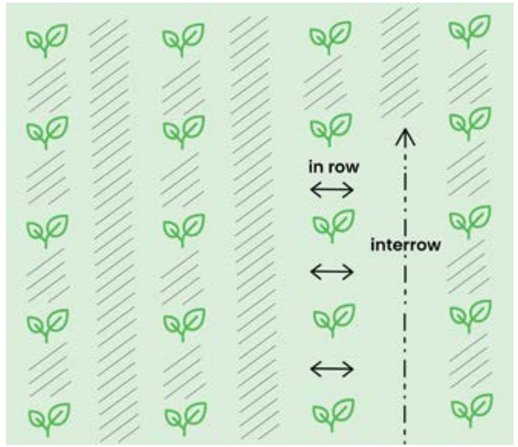
Although we began our journey with a focus on seeding and weeding sugar beets, our vision has always been grander.

We believe that precision farming is the future for a wide range of crops, and our commitment has driven us to constantly evolve and adapt. In close partnership with our valued customers and dedicated distributors, we've expanded the FarmDroid FD20's capabilities beyond its initial design.

The results speak for themselves. From our initial focus on sugar beet to the wide array of crops today, FarmDroid confidently ensures precision in both seeding and weeding.

+Weed

FarmDroid's ability to start blind mechanical weeding even before crops emerge is an unrivaled feature in the industry that sets it apart from the competition. While systems relying on camera technology require visible crops to differentiate them from weeds, FarmDroid FD20 knows the exact position of each seed it has sown, and therefore can perform mechanical weeding right after seeding. This capability keeps fields weed-free from the start, significantly reducing the need for labor-intensive and costly manual weeding. With fewer weeds, farmers can achieve healthier crop growth, higher yields, and lower operational costs.





Two-Way Weeding: Between the Plants and in Between the Rows

The FD20 employs two weeding wires per row, effectively targetting weeds in between the rows. For those stubborn intra-row weeds, an electrically operated weeding knife dynamically adjusts its timing based on the seeding distance, ensuring that weeds between the plants are also removed. While this system is optimized for general use, we recognize that every farm is unique, so the system can easily be adjusted to meet specific needs.



94% REDUCTION IN PLANT PROTECTION SUBSTANCES

+Spray

FarmDroid +Spray offers conventional farmers a unique advantage by combining mechanical weeding and micro spraying. The FarmDroid FD20 equipped with the +Spray system fights herbicide resistance while helping farmers adapt to increasingly strict regulations on plant protection substances.

Like the +Seed and +Weed systems, the FarmDroid +Spray system uses RTK GPS technology to apply plant protection substances in micro doses directly onto plants or the soil around them. This method reduces chemical usage by up to 94%, promoting healthier crops, cutting operational costs, and benefiting the environment.

The FarmDroid +Spray system can be used for both micro-spot spraying and band spraying. +Spray has a tank capacity to operate approximately 24 hours daily when spot spraying and 12 hours when band spraying.

Working fully autonomously, the +Spray system continuously monitors the spraying process, alerting farmers via notifications on the FarmDroid app, such as low liquid levels that require refilling. This allows for better planning and gives the farmers the freedom to work with a self-functioning system.

The FarmDroid +Spray features innovative PWM (Pulse Width Modulation) electric nozzles that apply minimal dosages with exceptional accuracy, ensuring that the exact amount of product is micro-sprayed where needed. FarmDroid +Spray automatically flushes the nozzles with clean water when in standstill mode or after spraying, preventing substances from drying out in the nozzles.







FarmDroid Benefits

- Reduced costs on multiple levels
- Green energy and CO_2 neutral operation
- Higher yield potential
- Payback period as low as two years
- Healthier crops
- Preserving soil structure and fertility
- Compatible with seeding and weeding more than 50 crops
- Exceptional 18 hours daily support during high season

FarmDroid for Organic Farmers: Harness the power of precision farming to grow healthier, organic crops

FarmDroid +Seed +Weed

FarmDroid is an ideal solution for organic farmers, offering a sustainable and efficient way to manage fields. The two most relevant functions for organic farming are FarmDroid +Seed for precision seeding and +Weed for mechanical weeding.

Using 8mm GPS precision, the FarmDroid FD20 sows seeds at exact intervals and optimal depths, ensuring perfect crop spacing for better crop growth and yield. Even before the crops sprout, the FD20's two-way mechanical weeding system removes weeds by targeting the spaces both between plants and between rows. This process drastically reduces the hours spent on manual weeding—one of the most labor-intensive tasks in organic farming! With the FarmDroid FD20, organic farmers can maintain clean fields, improve crop yields, and focus on what matters most: growing healthy, sustainable produce.



Success Stories of Organic Farmers



Germany

“The robot makes you less dependent on manual labor”

Anton Maier has been using FarmDroid to manage his organic sugar beet fields. With a shortage of manual labor, it was essential to minimize labor requirements, and FarmDroid proved to be an effective solution. Anton doesn’t have to monitor the robot regularly, only making occasional adjustments to optimize performance based on field conditions. He appreciates how reliably the robot operates with such minimal supervision. Anton highly values the support he received from the FarmDroid Care team, as it allowed him to quickly resolve his questions and optimized his experience.



Denmark

“Easy to use and very precise!”

Anders Tornslev Bach and his team were facing low yields and high expenses from traditional, manual weeding methods in their sugar beet fields. After learning about FarmDroid from a neighbor, they invested in the robot, and it transformed their operations. With FarmDroid, they minimized manual weeding substantially, reducing costs by 80%. Anders appreciates the control he now has over field management without relying on external help. For him, FarmDroid’s precision seeding, blind weeding, and solar-powered operation all stand out as key advantages, enabling healthy, clean plants at his farm without the need for fuel.

FarmDroid for Conventional Farmers: Optimize your results through high precision seeding and hybrid weeding

FarmDroid +Seed +Weed +Spray

As a conventional farmer, you may face challenges like herbicide resistance, high costs of plant protection substances, and stricter regulations on their use. To address these issues, we have developed a hybrid solution that enables you to stay ahead of the curve with our next-generation technology.

Like organic farmers, conventional farmers can utilize the FarmDroid +Seed system for high-precision seeding and the +Weed system for mechanical weeding. However, what distinguishes the FarmDroid FD20 for conventional farmers is its micro-spraying system, +Spray.

The FD20’s hybrid weeding system combines 94% mechanical weeding with just 6% spraying, helping you tackle the challenges mentioned above. After sowing seeds with 8mm GPS accuracy, the robot performs mechanical weeding several times throughout the season, targeting most weeds around the crops, even before they germinate. For any remaining weeds or specific crop care, the FD20’s +Spray function applies minimal amounts of plant protection substances directly onto the plants or around them, saving you up to 94% on material costs. This method promotes healthier crops and soil while allowing you to comply with regulations even before they take effect.



Success Stories of Conventional Farmers



The Netherlands

“A way to reduce chemicals is to remove weed with a robot”

For Joost Derks, one of the biggest challenges in growing chicory and sugar beets is managing weeds, which traditionally requires heavy use of chemicals. Wanting to drastically reduce chemical use, he found the right solution in the FarmDroid FD20 robot, which he now uses on 10 hectares of sugar beets and 12 hectares of chicory. Joost believes the robot is an impressive machine and encourages those interested in buying a FarmDroid to connect with farmers who already use one.



Denmark

“It is the future: We use less chemicals but we keep the field clean”

Niels Kamper experienced FarmDroid’s +Spray micro-spraying technology firsthand, reducing his use of plant protection substances by over 90% while achieving optimal plant growth! He is pleasantly surprised by FarmDroid’s ability to perform so well despite the variations in soil types and levels in Falster, Denmark.

FarmDroid FD20 Features



- Solar powered
- 8mm high precision RTK GPS
- Automatic operation all day
- Connected via FarmDroid App
- Approx. 1250kg lightweight robot
- Up to 12 active rows
- 3.5 meters working width
- Row-spacing 22.5cm-90cm

One Platform Three Robots: Navigate Hilly Terrains and Narrow Rows with Custom Wheel Configurations

Every terrain and farming practice is unique. Recognizing this, the FarmDroid FD20 offers three distinct front wheel setups to cater to diverse agricultural needs. For all three configurations, the dual back wheels can be adjusted between 160-230 cm in 10 cm increments.

With these configurations, the FarmDroid ensures optimized performance tailored to your specific farming conditions. Select the wheel setup that best suits your terrain and experience how the robot enhances your farming operations.







1. Passive Front Wheel

The Passive Front Wheel is the default wheel configuration, providing stability and guidance to the robot on various farm terrains and crops.

This setup is **ideal for flat fields** and is particularly suited for crops with **even row counts** and **wider row spacing**, such as sugar beets with 45-50 cm row spacing.

2. Dual Front Wheel

The Dual Front Wheel configuration equips the FD20 with two front wheels, replacing the Passive Front Wheel. This setup can be chosen with a new robot or retrofitted to an existing one, with adjustable wheel spacing between 160-230 cm in 10 cm increments to suit your preferences.

Typically used for **beds and flat fields**, the Dual Front Wheel configuration is ideal for crops with **narrow row spacing**, such as onions down to 22.5 cm, and it accommodates both even and uneven row numbers.

3. Active Front Wheel

The Active Front Wheel equips the FD20 with an actuator that tilts the front wheel, enhancing its adaptability on hilly terrain. This tilting mechanism leverages gravity to reduce front-end impact and enhance efficiency. By monitoring the load on its two electric motors, the system tilts the front wheel in the opposite direction to evenly distribute weight across the back wheels, ensuring traction and stable performance.

This configuration is ideal for **hilly areas with slopes above 8%** and **side inclines over 5%**, and it is suited for crops with **even row numbers** and **wider row spacing**, around 45-50 cm.

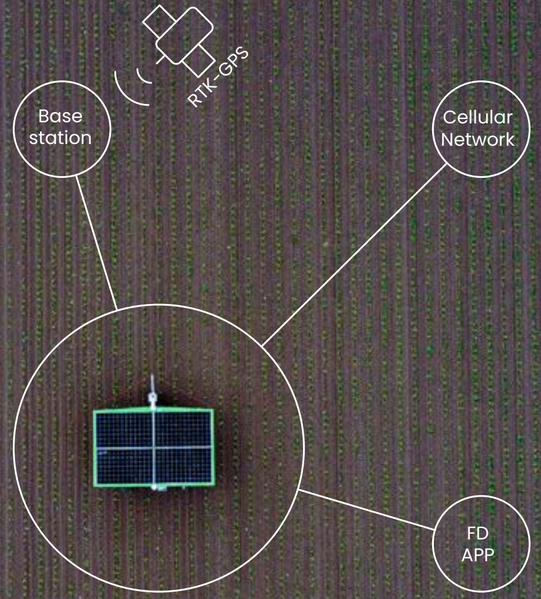


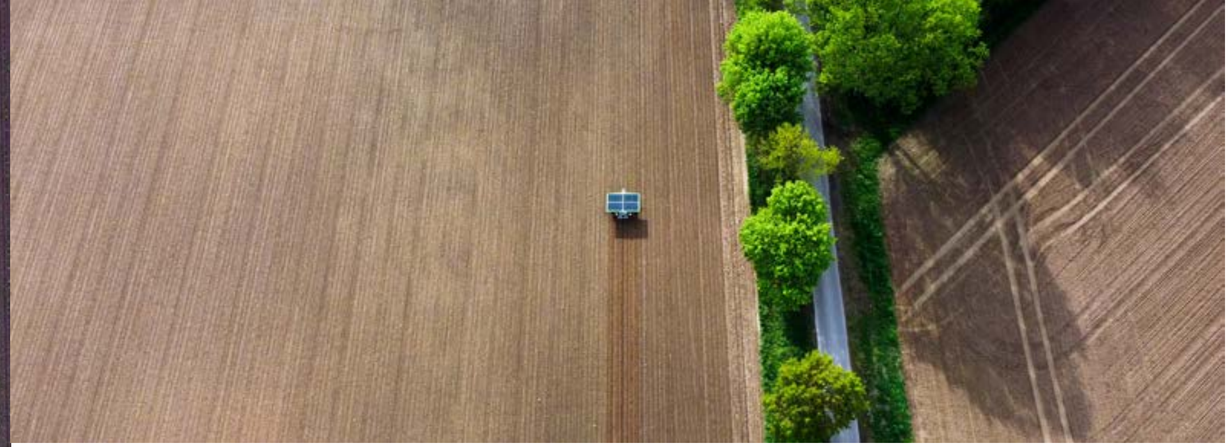
Unmatched 8mm Precision in Seeding, Weeding, and Spraying

The FarmDroid FD20 delivers unmatched navigational accuracy down to 8mm, thanks to its high-precision RTK GPS system connected to a dedicated base station. This ensures its operations remain seamlessly precise to the millimeter, setting a new standard in robotic farming.

Offering such precise accuracy requires the robot to operate at a slow and steady speed of 450-950 m/h, allowing it to optimally sow seeds and mechanically weed around them immediately after seeding. This "blind weeding" capability allows farmers to maintain clean fields from the very start, ensuring efficient and effective weed management.

The FarmDroid FD20's precision doesn't stop at seeding and mechanical weeding. With its retrofittable +Spray system, the robot applies plant protection substances with the same level of accuracy, micro-spraying directly onto plants or the soil around them. This precision reduces chemical use by up to 94%, ensuring targeted weed control while promoting healthier crops and protecting the environment.





Consistent Field Operations All Season: Powered by the Sun and Enhanced with a Power Bank

The FarmDroid FD20 epitomizes the future of sustainable farming, drawing all its operational energy directly from the sun. Equipped with four solar panels, the FD20 efficiently converts sunlight into power for its two batteries, with a maximum output of 1.6 kWh. Depending on weather conditions and operational demands, this energy storage allows for 18-24 hours of continuous work.

At a steady pace of 950 meters per hour, the FD20 covers approx. 5 hectares a day, and is recommended for fields up to 30 hectares per season. Powered by the sun, the robot eliminates the need for constant refueling or recharging—no more trips back and forth for energy top-ups. Your field becomes the FD20's home for the entire season, allowing you to focus on more important tasks.

For locations with frequent cloudy days, the FD20's runtime can be extended with power banks, which can be charged via solar panels or even at home.

Easy Guided Setup with the Field Setup Tool

Preparing your field is simple with our Field Setup Tool. Its lightweight, ergonomic design makes it easy to carry around, and its battery-powered functionality enhances portability. The Field Setup Tool pairs seamlessly with your smartphone and connects to the FarmDroid App, guiding you through mapping the corners and obstacles of your field.



1

Take the Field Setup tool to your field



3

Record corner points and obstacles



5

Start seeding with the FarmDroid



2

Connect to your FarmDroid App



4

Send the recorded field to your FarmDroid



Stay Connected to Your Robot Anytime, Anywhere with the FarmDroid App

Experience the future of farming management with the FarmDroid App. This mobile application serves as your virtual window to the FarmDroid FD20, keeping you informed and in control, no matter where you are.

With the FarmDroid App, you can track your robot's location and receive updates via email, SMS, or push notifications tailored to your preferences. Need to troubleshoot or check on operations? Skip the drive to the field—just tap the app to access your robot's live camera feed and monitor its progress and current tasks.

The app also allows you to set up your field directly from your phone using the Field Setup Tool and connect with the FarmDroid Care support team in the unlikely event of an operational hiccup.

Instant notifications

Easy field setup

Connect to FarmDroid Care

Live video feed

Live field progress

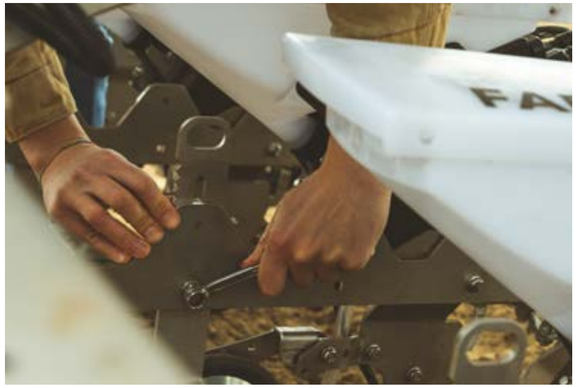
FarmDroid Care: Guiding Your Robotic Farming Journey

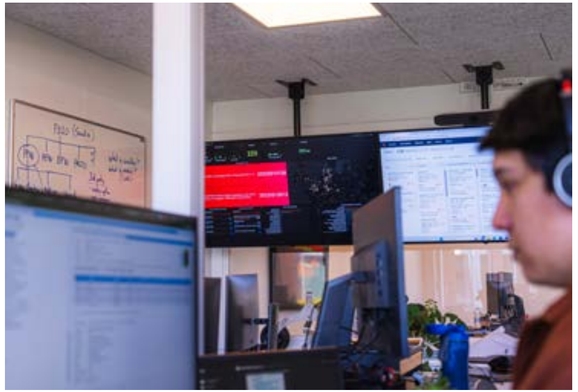
Transitioning from a tractor to a robot may sound complex and daunting, but our FarmDroid Care team ensures a smooth and successful experience. When you reach out to us, you'll be greeted by dedicated experts ready to support your agricultural success. With FarmDroid, you're never alone in the field; help is just a click away!

Our FarmDroid Care team can remotely access all robot systems, providing quick assistance to keep your operations running smoothly. With service plans designed for peace of mind, we ensure your farm thrives throughout the season.

Our team is a diverse group from all over the world, united by one goal: supporting our FarmDroid users. We're more than just a support team—we're your partners in farming success!

Are you considering transitioning to robot technology? Take on the challenge with confidence, because FarmDroid Care is your reliable support, always by your side.





How It Works: From Delivery to Operation



1

Delivery
Unpack the robot and install the GPS base station within 10 km from the field(s).



2

Field setup
Mark the field's corner points and obstacles in the field.



3

Seeding
Fill the seeds in the seed containers, set the desired seed depth and distance. You're ready to start seeding.



4

Blind weeding
Perform blind weeding right after seeding, before the crop germinates. By doing so, you let the plants germinate without competition from weeds.



5

Mechanical weeding
Adjust the robot's weeding proximity to the plants and initiate weeding both within the row and between the plants.



6

Micro spraying
If you are a conventional farmer you can spray liquid plant protection substances in micro doses onto the plants.

FARMDROID

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